
RESEARCH BRIEF — FOR CAREGIVERS

Cognitive Engagement

The value of mentally stimulating activities in later life — and
how caregivers can support them

A research brief from mynaani — September 2026

Executive summary

For most of life, staying mentally active is a choice. In later life, it becomes a protective habit. A growing body of longitudinal research shows that older adults who regularly engage in mentally stimulating activities maintain cognitive function longer and are at lower risk of mild cognitive impairment and dementia than those who do not. The benefit is not limited to crossword puzzles or reading. It extends to learning new skills, using a computer, playing games, making things with the hands, and participating in meaningful conversations.

This brief synthesizes the evidence for cognitive engagement as a modifiable protective factor and explains the caregiver's role in supporting it. Caregivers are not clinicians, and they cannot replace structured therapy. But they are uniquely positioned to notice, encourage, and remove the friction that stops an older adult from staying mentally active. The research shows that caregiver-delivered cognitive stimulation, when simple and consistent, can improve cognition, quality of life, and even caregivers' own sense of competence.

The conclusion is practical: keeping an older adult mentally engaged is one of the few protective actions available at home, at low cost, with evidence behind it.

1. The idea: use it or lose it

The "use it or lose it" hypothesis is the starting point for most research on cognitive aging. It suggests that mental activity strengthens the neural and cognitive systems that support memory, reasoning, and problem-solving. The brain does not simply wear out with age; it adapts to what is asked of it. When the demands stop, the capacity to meet them declines faster.

Cognitive reserve is the related concept. It refers to the brain's ability to tolerate age-related change or pathology by drawing on a richer network of skills and connections. Education, occupation, and lifelong learning build

reserve. Later-life cognitive activity helps maintain it. Reserve does not prevent dementia, but it can delay the appearance of symptoms and preserve day-to-day function.

For family caregivers, this means that mental engagement is not a luxury. It is one of the few protective behaviors that can be supported at home.

2. What counts as mentally stimulating

The research literature uses several overlapping categories. They are not mutually exclusive, and the strongest evidence points to variety and regularity rather than any single activity.

Adult literacy activities. Writing letters or journaling, using a computer, and taking education classes. These are consistently associated with slower cognitive decline in large cohort studies.

Active mental activities. Playing games, cards, chess, or doing crosswords and puzzles. These require planning, rule-following, and working memory.

Creative or craft activities. Woodwork, metalwork, painting, drawing, knitting, quilting. These combine fine motor control with problem-solving and often produce a tangible result.

Social and intellectual leisure. Reading, visiting museums, attending lectures, meaningful conversation. These activities combine mental effort with social connection.

Learning new technology. Using a computer, the internet, or a smartphone for email, video calls, or information-seeking. Several large studies have found computer use to be independently protective.

The common thread is not difficulty. It is sustained attention, meaningful goal, and novelty. Activities that are too easy or too repetitive stop providing benefit.

3. What the research shows

The evidence comes from large longitudinal studies in several countries. It is consistent on direction, even when the exact numbers differ.

Cognitive activity predicts better cognitive function over time. A study of 1,168 community-dwelling older adults, primarily Black women, followed for an average of 6.4 years found that more frequent cognitive activity was associated with better cognitive functioning at baseline and over time, even after accounting for biomarkers of neurodegeneration (PMC, 2024).

Mental activity is linked to lower dementia risk. A 10-year prospective cohort of 10,318 Australian adults aged 70 and older found that more frequent engagement in adult literacy activities was associated with an 11% lower risk of dementia, and active mental activities with a 9% lower risk. Creative and passive mental activities were also protective, though to a lesser extent. Social outings and interpersonal networks were not associated with risk in this sample, suggesting that the *type* of engagement matters (JAMA Network Open, 2023).

More activities, more protection. The Mayo Clinic Study of Aging followed 2,000 cognitively unimpaired adults for a median of five years. Engaging in a higher number of mentally stimulating activities in late life was associated with a substantially lower risk of incident mild cognitive impairment. Compared with one or no activities, any two activities reduced risk by 28%, any three by 45%, any four by 56%, and all five by 43% (Neurology, 2019).

Computer use stands out. The same Mayo Clinic study found that using a computer was associated with a decreased risk of MCI regardless of when people started. Those who used a computer in both midlife and late life had a 37% lower risk; those who started in late life had a 30% lower risk.

Intellectual activity is the strongest leisure domain. A longitudinal analysis of 6,847 Chinese older adults followed for 6.1 years found that higher baseline frequency of intellectual, social, and physical activities were each associated with slower cognitive decline. Intellectual activity showed the strongest association, and engaging in two or three domains together produced a larger benefit than any single domain (BMC Geriatrics, 2024).

Increases in activity matter, even without a college education. A study of 2,130 U.S. adults followed for approximately nine years found that increasing cognitive activity over time was associated with reduced decline in executive functioning, especially among those without a college degree.

This suggests that the habit itself, not just prior education, is protective (Frontiers in Psychiatry, 2021).

Mind-stimulating leisure is associated with better health and longevity outcomes. A six-year prospective study of 19,821 middle-aged and older adults across 15 European countries found that reading almost daily was associated with lower risk of depression, pain, daily functioning limitations, cognitive impairment, and loneliness. Serious solitary leisure activities were associated with lower risk of death by any cause and lower risk of depression (Frontiers in Public Health, 2023).

4. The caregiver's role

Caregivers do not have to design an intervention. They can, however, create the conditions for engagement. The evidence on caregiver-delivered cognitive stimulation is especially relevant.

Caregivers can deliver effective cognitive stimulation at home. A randomized controlled trial of individual cognitive stimulation therapy (iCST) delivered by family caregivers found that 30-minute sessions, two or three times a week, over 25 weeks, were associated with improved cognition and quality of life for people with dementia. Caregivers reported improved caregiving confidence and a better relationship with the person they supported (PLOS Medicine, 2016).

A shorter program can also help. A Portuguese trial of a home-based individual cognitive stimulation program delivered by caregivers to people with cognitive impairment showed a significant improvement in orientation and ability to follow commands. The program ran for 12 weeks, three times a week. Caregivers reported that they noticed improvements in functioning as the sessions progressed (Frontiers in Psychology, 2021).

Training improves outcomes for both the older adult and the caregiver. A quasi-experimental study of 160 older adults found that those cared for by caregivers who had completed a three-month cognitive stimulation training programme showed gains in quality of life and cognitive ability, and maintained functional ability. Older adults supported by *family* caregivers improved on more cognitive sub-items than those supported by professional

caregivers, suggesting that the emotional bond and everyday proximity of family support can enhance the benefit (PMC, 2022).

Caregivers influence activity diversity. A study of 6,275 informal caregivers and 4,271 non-caregivers using the U.S. Health and Retirement Study found that caregivers reported more frequent and more diverse activity engagement than non-caregivers. Higher activity frequency and diversity were associated with better cognition and memory, though the benefit was somewhat attenuated among caregivers, likely because of the competing demands of caregiving itself (Research on Aging, 2024).

The practical message is that a caregiver's role is not to become a therapist. It is to introduce, model, and sustain mentally stimulating activities that fit the older adult's interests and abilities.

5. How to support cognitive engagement at home

The research suggests a few practical principles:

- **Match the activity to the person, not the diagnosis.** Puzzles help some people; others respond to music, crafts, conversation, or learning to use a new device. The activity must hold meaning for the older adult.
- **Keep it regular but brief.** Most effective caregiver-delivered programs use short, frequent sessions — 20 to 30 minutes, several times a week. Longer is not necessarily better.
- **Start from ability, not deficit.** Build on what the person can still do and enjoys. Nothing undermines engagement faster than a task that feels like a test.
- **Remove friction.** Locate the activity where the person already sits. Keep materials visible. Reduce the number of steps needed to begin. Friction is the enemy of habit.
- **Invite, do not instruct.** Autonomy predicts participation. The caregiver can suggest, but the older adult should choose.
- **Make it social when possible.** Conversation, shared games, or intergenerational contact add social and emotional value to the mental effort.

- **Track mood, not just performance.** Enjoyment and sense of purpose are part of why activities are protective. A good session is one the older adult wants to repeat.

6. Conclusion and next step

Mental activity is one of the few protective factors for cognitive aging that is available to almost everyone, regardless of income, diagnosis, or setting. The evidence is strong enough to act on: regular, varied, and meaningful cognitive engagement is associated with better function, lower dementia risk, and better quality of life. For family caregivers, supporting that engagement is a concrete, evidence-grounded way to help.

Mynaani's AI learning program is designed to be one such activity — a calm, self-paced, mentally engaging introduction to artificial intelligence for adults 55 and older. To learn more, or to request this brief for your family or organization, write to **hello@mynaani.com**.

About mynaani

Mynaani is an AI learning program designed specifically for adults 55+, built on geragogical principles and a patent-pending cognitively-protective interface. We have an interest in the conclusion of this paper; readers should weigh it accordingly. Every statistic cited is third-party and verifiable below.

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